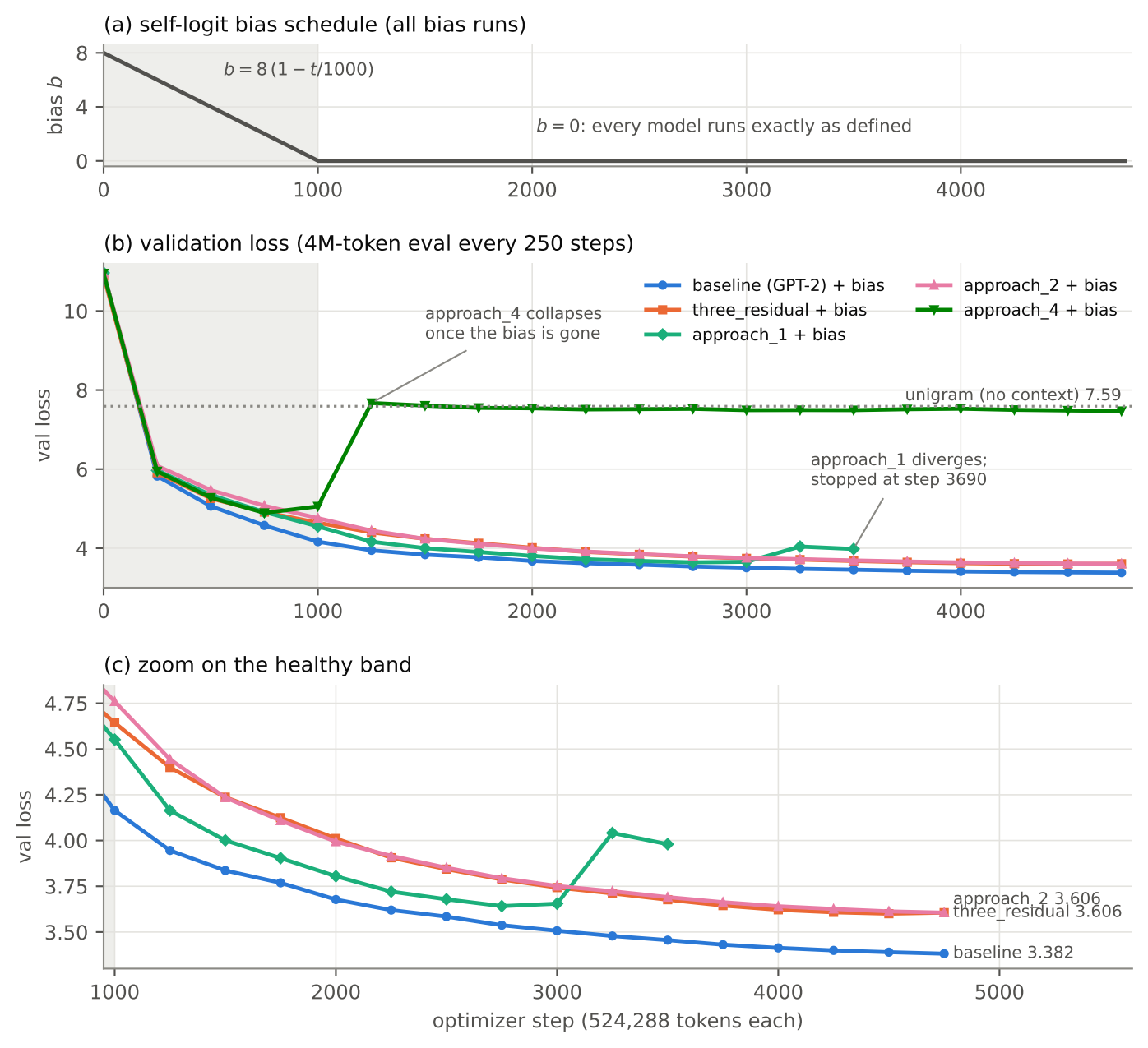
(a) self-logit bias schedule (all bias runs)
bias b
b = 8 (1 − t/1000)
b = 0: every model runs exactly as defined
(b) validation loss (4M-token eval every 250 steps)
val loss
baseline (GPT-2) + bias
three_residual + bias
approach_1 + bias
approach_2 + bias
approach_4 + bias
approach_4 collapses
once the bias is gone
unigram (no context) 7.59
approach_1 diverges;
stopped at step 3690
(c) zoom on the healthy band
val loss
approach_2 3.606
three_residual 3.606
baseline 3.382
optimizer step (524,288 tokens each)